# Assignment - 2

1) 1.1)

$$p(\theta|y) = \frac{L(\theta|y)\,p(\theta)}{\int L(\theta|y)\,p(\theta)\,d\theta}$$

$$= \frac{\frac{10!}{7! \times 3!}\,\theta^7(1-\theta)^3 \quad (1)}{\frac{1}{11}}$$

$$= \left(\frac{11 \times 10!}{7! \times 3!}\right)\theta^7\,(1-\theta)^3$$

$$= 1320\,\theta^7\,(1-\theta)^3$$

(a) $\theta = 0.75$

$$p(0.75|y) = 1320 \times (0.75)^7 \times (0.25)^3$$
$$= 2.752$$

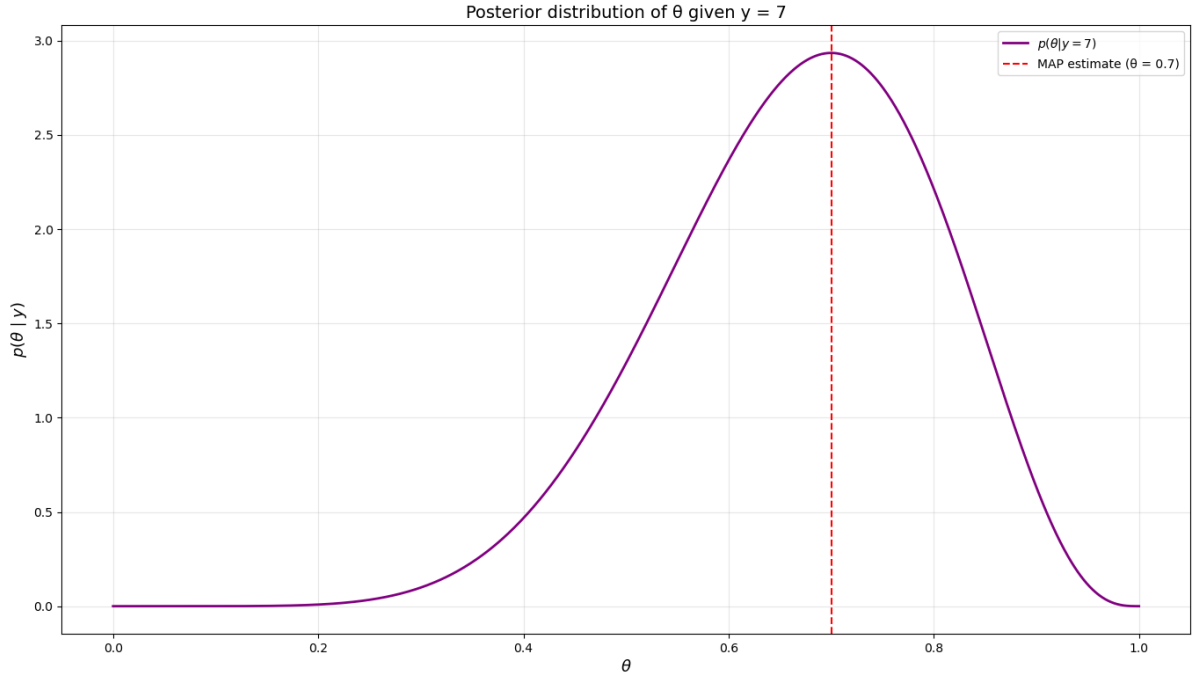
(b) $\theta = 0.25$

$$p(0.25|y) = 1320 \times (0.25)^7 \times (0.25)^3$$
$$= 0.034$$

(c) $\theta = 1$

$$p(1|y) = 1320 \times (1)^7\,(0)^3 = 0$$

1.2)




1.3) $p(\theta|y) = 1320\ \theta^7 (1-\theta)^3$

so, for maximizing this
lets take natural log since it is an increasing function

$$\ln(1320) + 7\ln\theta + 3\ln(1-\theta)$$

then, differentiate it wrt $\theta$ and equate to 0.

$$= 0 + \frac{7}{\theta} + \frac{3}{(1-\theta)}(-1) = 0$$

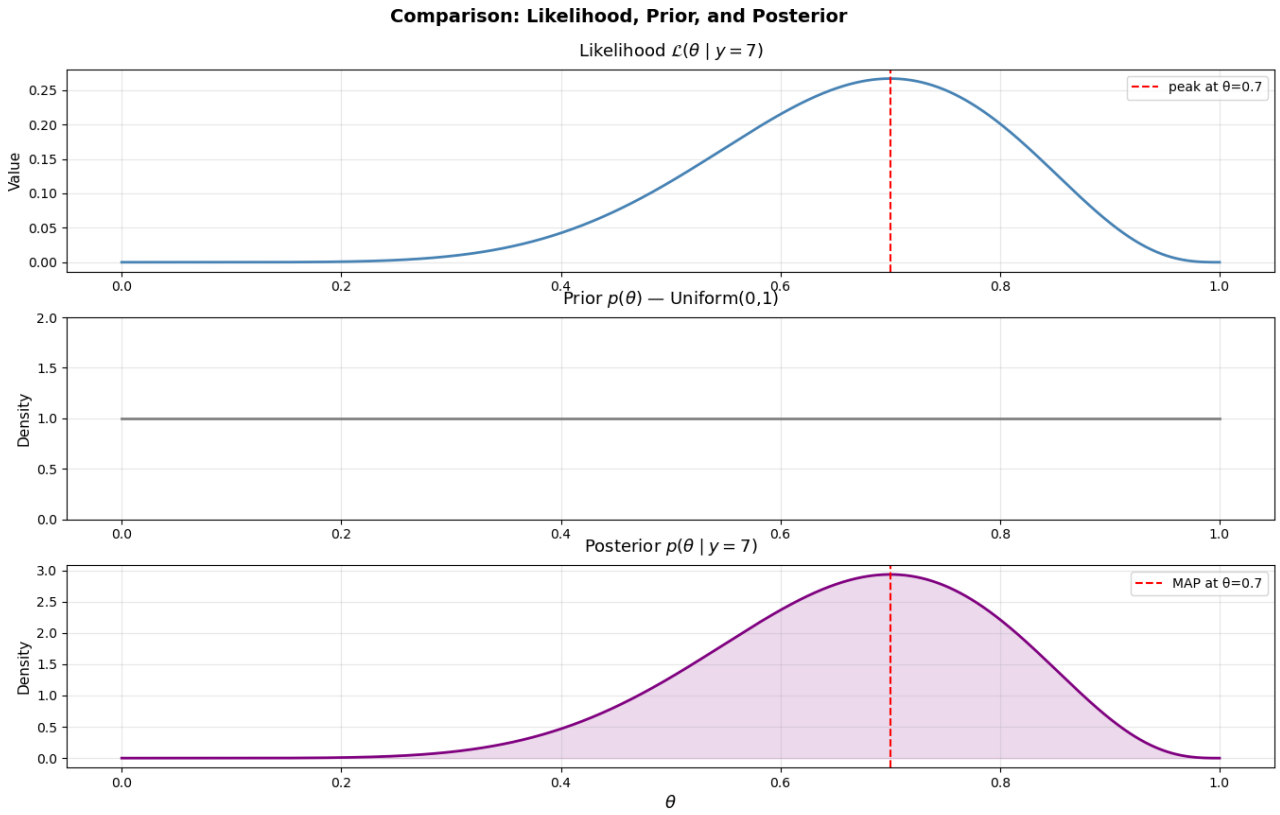
$$\Rightarrow \frac{7}{\theta} = \frac{3}{1-\theta}$$

$$7 - 7\theta = 3\theta$$

$$10\theta = 7$$

$$\theta = \frac{7}{10} = 0.7$$

1.4)




2) 2.1) $p'(\mu|\sigma,y) = L(\mu,\sigma|y)\, p(\mu)$

$$= \frac{1}{(\sigma\sqrt{2\pi})^n} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i-\mu)^2} \times \frac{1}{25\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\mu-250}{25}\right)^2}$$

$$= \frac{1}{(50\sqrt{2\pi})^8} e^{\frac{-1}{2(50)^2}\sum_{i=1}^{8}(y_i-\mu)^2} \times \frac{1}{25\sqrt{2\pi}} e^{\frac{-1}{2}(25)^2(\mu-250)^2}$$

for $\mu = 300$

$$\sum (y_i - 300)^2 = 0 + 900 + 8100 + 22500 + 40000 + 100 + 144400 + 22500 = 238500$$

$$= \frac{1}{(50\sqrt{2\pi})^8} e^{\frac{-238500}{5000}} \left( \frac{1}{25\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{300-250}{25}\right)^2} \right)$$

$$= 1.91 \times 10^{-22}$$

for $\mu = 900$

$$\sum (y_i - 900)^2 = 360000 + 396900 + 260100 + 202500 + 160000 + 372100 + 48400 + 202500 = 2002500$$

$$= \frac{1}{(50\sqrt{2\pi})^8} e^{\frac{-2002500}{5000}} \left( \frac{1}{25\sqrt{2\pi}} e^{\frac{-1}{2}\left(\frac{650}{25}\right)^2} \right)$$

$$\approx 0$$

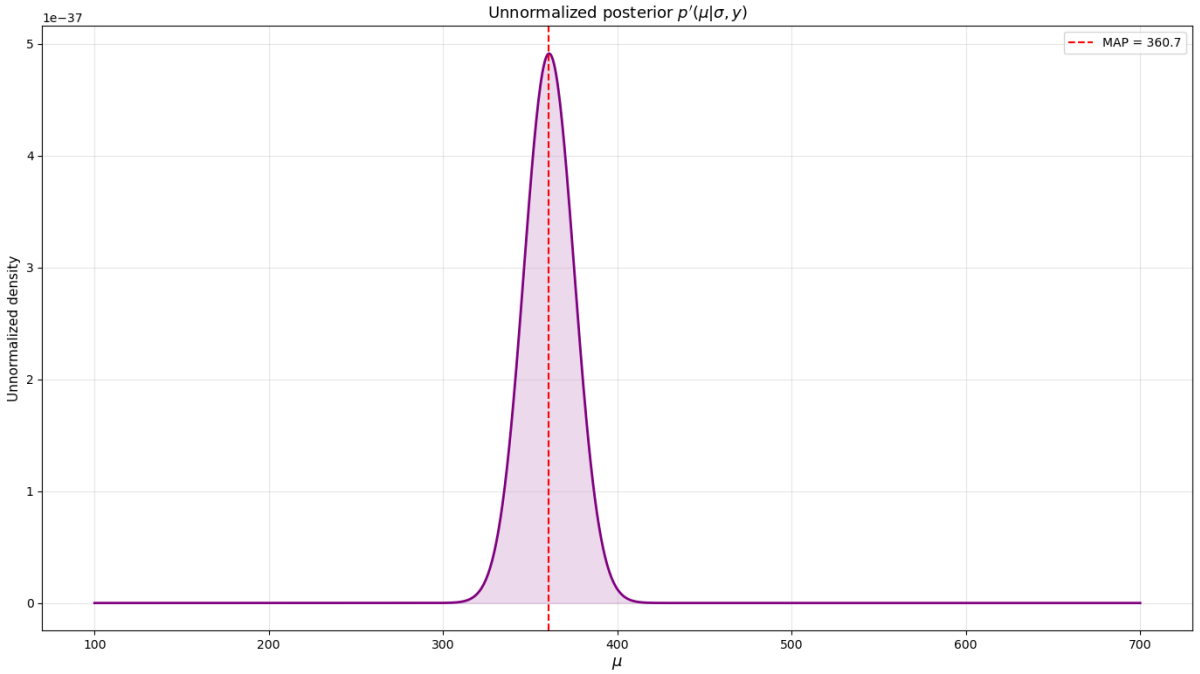
for $\mu = 50$

$$\sum (y_i - 50)^2 = 62500 + 48400 + 115600 + 160000 + 202500 + 57600 + 390400 + 160000 = 1197000$$

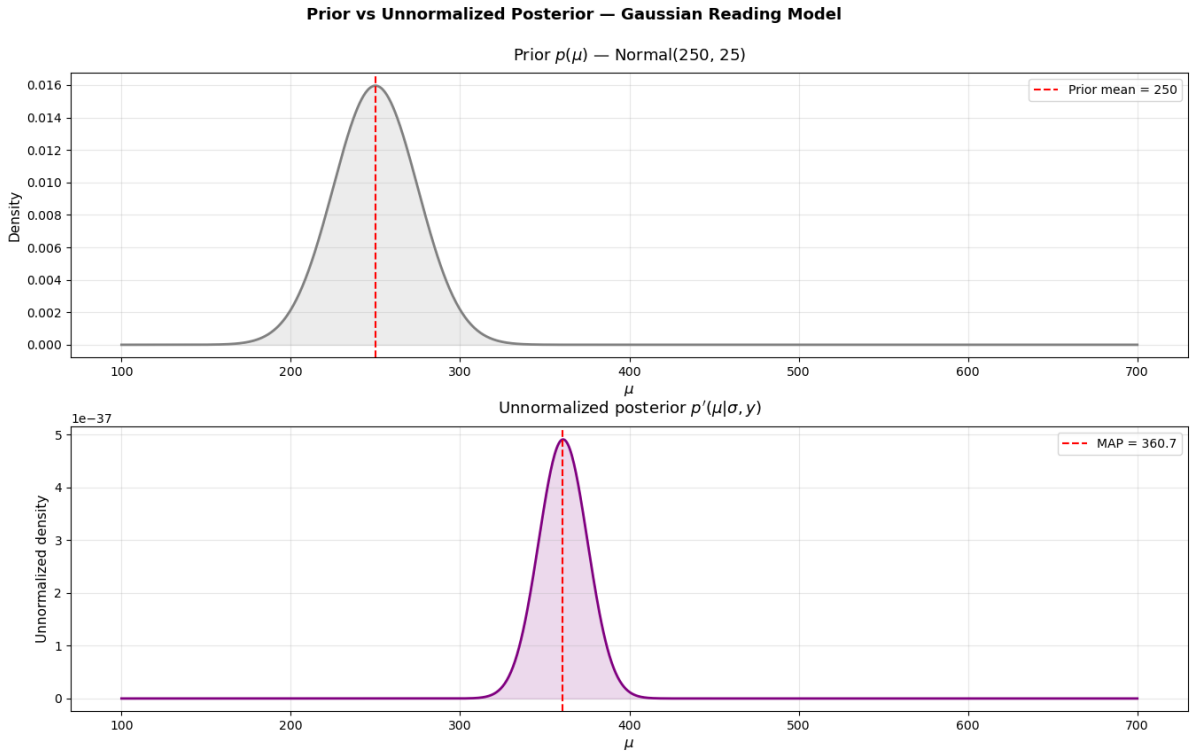
$$= \frac{1}{(50\sqrt{2\pi})^8} e^{\frac{-1197000}{5000}} \left( \frac{1}{25\sqrt{2\pi}} e^{\frac{-1}{2}\left(\frac{200}{25}\right)^2} \right)$$

$$\approx 0.$$

2.2)



2.3)




3) 3.1) So, the posterior is Gamma with $\alpha$ = previous $\alpha$ + data and $\beta$ = previous $\beta$ + no. of observations

| Day | Data, k | Prior | Posterior |
|---|---|---|---|
| 1 | 25 | Gamma(40, 2) | Gamma(65, 3) |
| 2 | 20 | Gamma(65, 3) | Gamma(85, 4) |
| 3 | 23 | Gamma(85, 4) | Gamma(108, 5) |
| 4 | 27 | Gamma(108, 5) | Gamma(135, 6) |

So, prior for day 5 is $\lambda \sim$ Gamma(135, 6)

3.2) Predicted number of road accidents is

$$E(k_5) = E(\lambda)$$

because

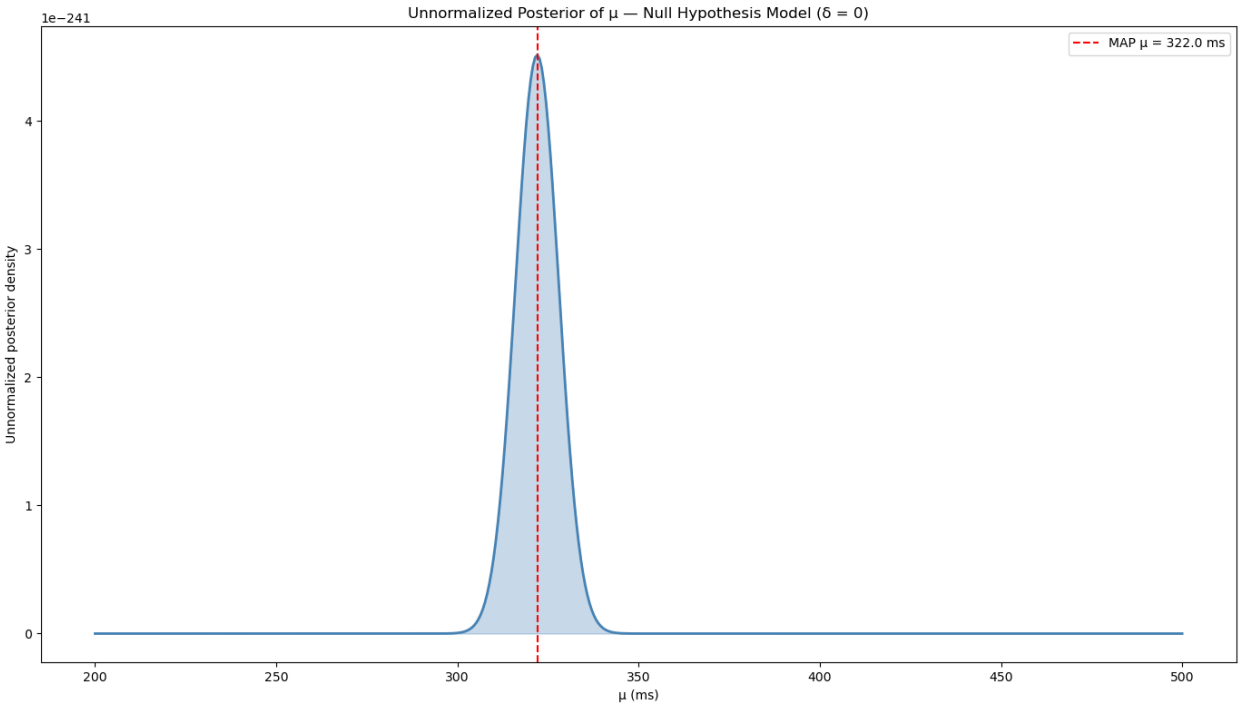
$$E(k_5|\lambda) = \lambda$$

$$E(\lambda) = \frac{\alpha}{\beta} = \frac{135}{6} = 22.5$$

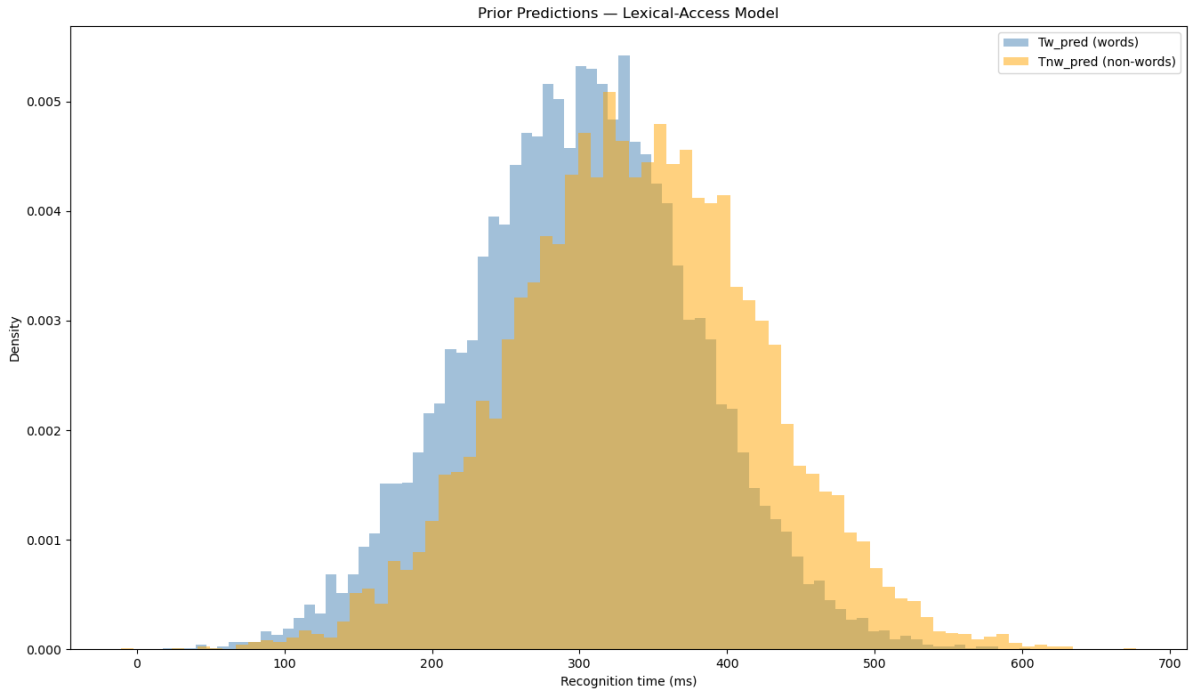
Suggesting that 22 or 23 accidents are predicted to happen on day 5.

4.5)

4.5.1)




4.5.2)



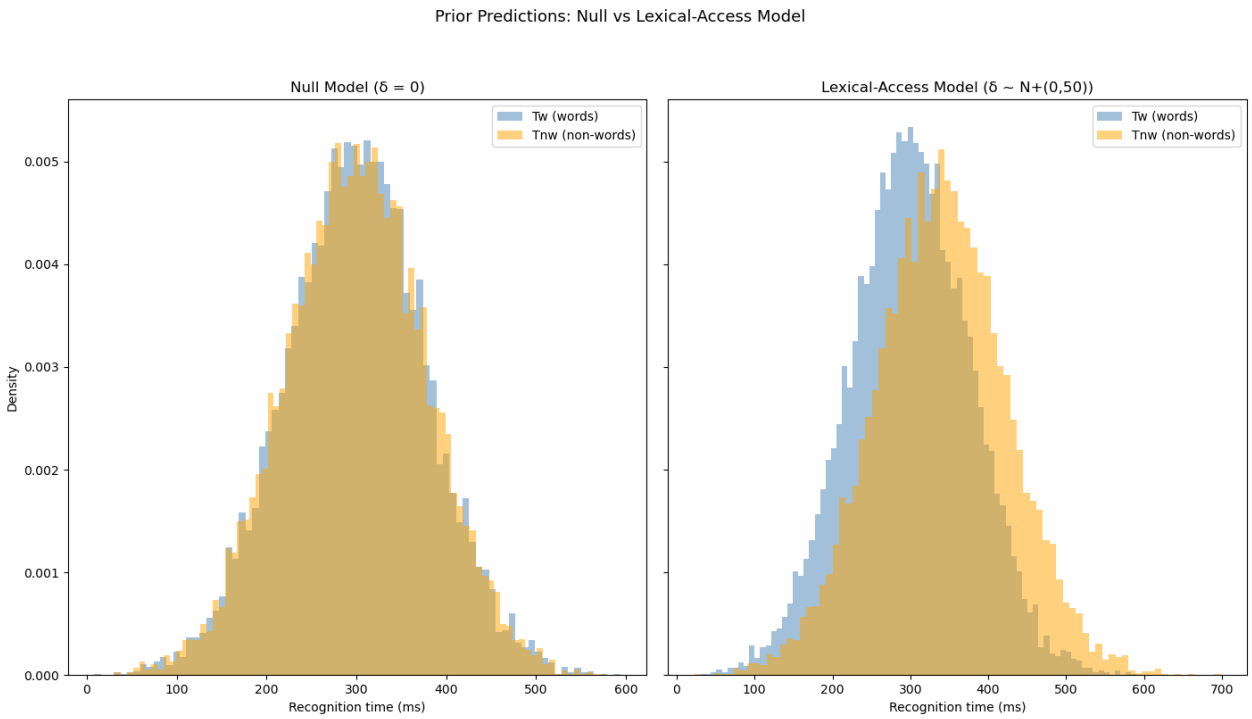

for 10000 samples of $\mu$ and $\sigma$

Mean Tw_pred = 300.78 ms

Mean Tnw_pred = 339.32 ms

Expected shift due to delta prior mean (mean of delta samples) = 40.31 ms

4.5.3)



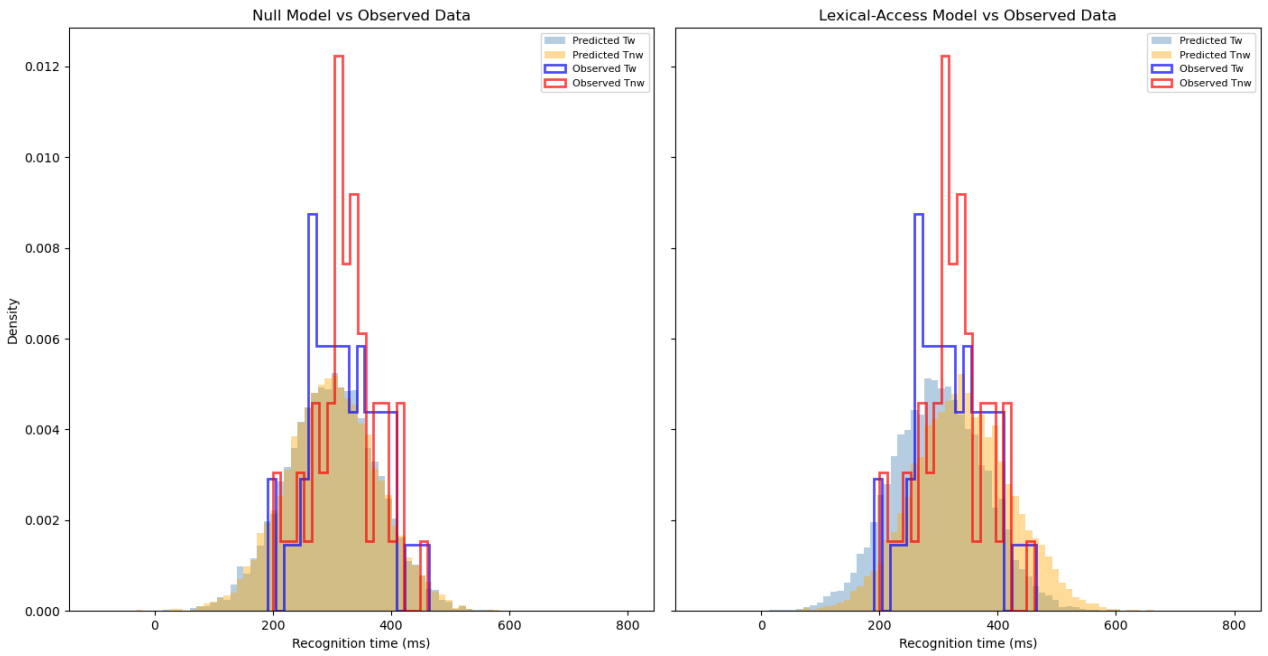

Null model

Mean Tw = 299.82 ms
Mean Tnw = 299.65 ms
Difference = −0.17 ms

Lexical-access model

Mean Tw = 299.90 ms
Mean Tnw = 339.82 ms
Difference = 39.92 ms

4.5.4)



Observed data

Mean $T_W$ = 321.37 ms

Mean $T_{NW}$ = 323.24 ms

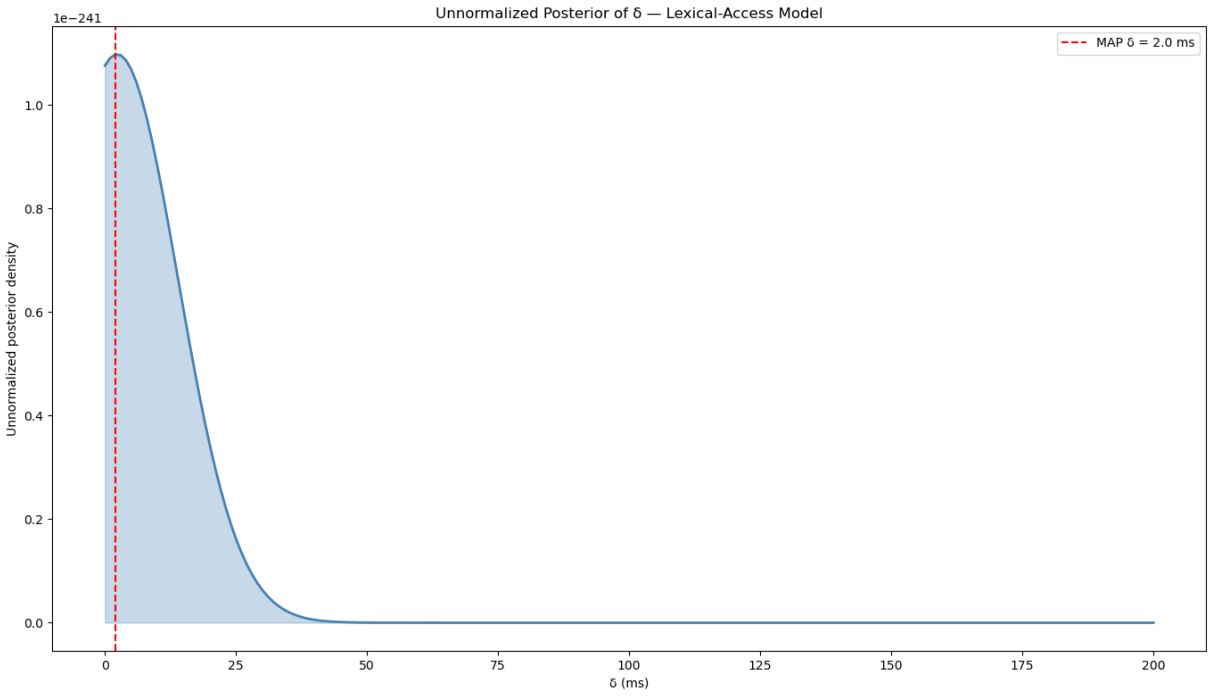
Observed difference = 1.86 ms

Lexical-access prior predicted difference ≈ 40ms

Null prior predicted difference ≈ 0ms

So, Null hypothesis model seems more consistent since 1.86 is closer to 0 than 40.

4.5.5)



```
import numpy as np
import matplotlib.pyplot as plt
from math import comb
from scipy.stats import norm
from scipy.stats import truncnorm
import pandas as pd
y = 7
n = 10
marginal_likelihood = 1/11
theta = np.linspace(0, 1, 1000)
coeff = comb(n, y)
posterior = (coeff * theta**y * (1 - theta)**(n - y)) / marginal_likelihood
plt.figure(figsize=(8, 5))
plt.plot(theta, posterior, color='purple', linewidth=2, label=r'$p(\theta|y=7)$')
plt.axvline(x=0.7, color='red', linestyle='--', linewidth=1.5, label='MAP
    estimate (θ = 0.7)')
plt.xlabel(r'$\theta$', fontsize=13)
plt.ylabel(r'$p(\theta \mid y)$', fontsize=13)
plt.title('Posterior distribution of θ given y = 7', fontsize=14)
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

likelihood = coeff * theta**y * (1 - theta)**(n - y)
prior = np.ones_like(theta)
posterior = likelihood / marginal_likelihood

fig, axes = plt.subplots(3, 1, figsize=(8, 12))


axes[0].plot(theta, likelihood, color='steelblue', linewidth=2)
axes[0].axvline(x=0.7, color='red', linestyle='--', linewidth=1.5, label='peak at
     θ=0.7')
axes[0].set_title(r'Likelihood $\mathcal{L}(\theta \mid y=7)$', fontsize=13, pad
    =10)
axes[0].set_ylabel('Value', fontsize=11)
axes[0].legend()
axes[0].grid(True, alpha=0.3)


axes[1].plot(theta, prior, color='gray', linewidth=2)
axes[1].set_title(r'Prior $p(\theta)$  Uniform(0,1)', fontsize=13, pad=10)
axes[1].set_ylabel('Density', fontsize=11)
axes[1].set_ylim(0, 2)
axes[1].grid(True, alpha=0.3)


axes[2].plot(theta, posterior, color='purple', linewidth=2)
axes[2].fill_between(theta, posterior, alpha=0.15, color='purple')
axes[2].axvline(x=0.7, color='red', linestyle='--', linewidth=1.5, label='MAP at
    θ=0.7')
axes[2].set_title(r'Posterior $p(\theta \mid y=7)$', fontsize=13, pad=10)
axes[2].set_ylabel('Density', fontsize=11)
axes[2].set_xlabel(r'$\theta$', fontsize=13)
axes[2].legend()
axes[2].grid(True, alpha=0.3)
```

```
plt.suptitle('Comparison: Likelihood, Prior, and Posterior', fontsize=14,
    fontweight='bold')
plt.tight_layout(rect=[0, 0, 1, 0.97])
plt.show()


y = np.array([300, 270, 390, 450, 500, 290, 680, 450])
n = len(y)
sigma = 50
prior_mean = 250
prior_sd = 25


mu = np.linspace(100, 700, 1000)


likelihood = np.array([
    (1 / (sigma * np.sqrt(2 * np.pi))**n) * np.exp(-np.sum((y - m)**2) / (2 *
    sigma**2))
    for m in mu
])


prior_vals = norm.pdf(mu, loc=prior_mean, scale=prior_sd)


unnorm_posterior = likelihood * prior_vals


map_mu = mu[np.argmax(unnorm_posterior)]


plt.figure(figsize=(8, 5))
plt.plot(mu, unnorm_posterior, color='purple', linewidth=2)
plt.fill_between(mu, unnorm_posterior, alpha=0.15, color='purple')
plt.axvline(x=map_mu, color='red', linestyle='--', linewidth=1.5, label=f'MAP = {
    map_mu:.1f}')
plt.xlabel(r'$\mu$', fontsize=13)
plt.ylabel("Unnormalized density", fontsize=11)
plt.title(r"Unnormalized posterior $p'(\mu|\sigma,y)$", fontsize=13)
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

fig, axes = plt.subplots(2, 1, figsize=(8, 10))


axes[0].plot(mu, prior_vals, color='gray', linewidth=2)
axes[0].fill_between(mu, prior_vals, alpha=0.15, color='gray')
axes[0].axvline(x=prior_mean, color='red', linestyle='--', linewidth=1.5, label='
    Prior mean = 250')
axes[0].set_title(r'Prior $p(\mu)$  Normal(250, 25)', fontsize=13, pad=10)
axes[0].set_ylabel('Density', fontsize=11)
axes[0].set_xlabel(r'$\mu$', fontsize=12)
axes[0].legend()
```

```
axes[0].grid(True, alpha=0.3)


axes[1].plot(mu, unnorm_posterior, color='purple', linewidth=2)
axes[1].fill_between(mu, unnorm_posterior, alpha=0.15, color='purple')
axes[1].axvline(x=map_mu, color='red', linestyle='--', linewidth=1.5, label=f'MAP
      = {map_mu:.1f}')
axes[1].set_title(r"Unnormalized posterior $p'(\mu|\sigma,y)$", fontsize=13, pad
     =10)
axes[1].set_ylabel('Unnormalized density', fontsize=11)
axes[1].set_xlabel(r'$\mu$', fontsize=12)
axes[1].legend()
axes[1].grid(True, alpha=0.3)

plt.suptitle('Prior vs Unnormalized Posterior  Gaussian Reading Model', fontsize
     =13, fontweight='bold')
plt.tight_layout(rect=[0, 0, 1, 0.97])
plt.show()




url = "https://raw.githubusercontent.com/yadavhimanshu059/CGS698C/main/notes/
     Module-2/recognition.csv"
dat = pd.read_csv(url, index_col=0)
Tw  = dat["Tw"].values
Tnw = dat["Tnw"].values

sigma = 60
delta = 0

mu_grid = np.linspace(200, 500, 500)

unnorm_posterior = []

for mu in mu_grid:
    L_Tw  = np.prod(norm.pdf(Tw,  loc=mu,           scale=sigma))
    L_Tnw = np.prod(norm.pdf(Tnw, loc=mu + delta,   scale=sigma))
    p_mu  = norm.pdf(mu, loc=300, scale=50)

    unnorm_posterior.append(L_Tw * L_Tnw * p_mu)

unnorm_posterior = np.array(unnorm_posterior)

map_mu = mu_grid[np.argmax(unnorm_posterior)]
print(f"MAP estimate of mu: {map_mu:.2f} ms")


plt.figure(figsize=(8, 5))
plt.plot(mu_grid, unnorm_posterior, color="steelblue", linewidth=2)
plt.fill_between(mu_grid, unnorm_posterior, alpha=0.3, color="steelblue")
plt.axvline(map_mu, color="red", linestyle="--", label=f"MAP μ = {map_mu:.1f} ms
     ")
plt.xlabel("μ (ms)")
plt.ylabel("Unnormalized posterior density")
```

```
plt.title("Unnormalized Posterior of μ  Null Hypothesis Model (δ = 0)")
plt.legend()
plt.tight_layout()
plt.show()



N = 10000


mu_samples    = np.random.normal(loc=300, scale=50, size=N)


a = (0 - 0) / 50
delta_samples = truncnorm.rvs(a=a, b=np.inf, loc=0, scale=50, size=N)

Tw_pred  = np.random.normal(loc=mu_samples,             scale=60)
Tnw_pred = np.random.normal(loc=mu_samples+delta_samples, scale=60)


plt.figure(figsize=(8, 5))
plt.hist(Tw_pred,  bins=80, alpha=0.5, color="steelblue", label="Tw_pred (words)"
    ,     density=True)
plt.hist(Tnw_pred, bins=80, alpha=0.5, color="orange",    label="Tnw_pred (non-
    words)", density=True)
plt.xlabel("Recognition time (ms)")
plt.ylabel("Density")
plt.title("Prior Predictions  Lexical-Access Model")
plt.legend()
plt.tight_layout()
plt.show()

print(f"Mean Tw_pred:  {Tw_pred.mean():.2f} ms")
print(f"Mean Tnw_pred: {Tnw_pred.mean():.2f} ms")
print(f"Expected shift due to delta prior mean: {delta_samples.mean():.2f} ms")

N = 10000
sigma = 60


mu_null      = np.random.normal(300, 50, N)
Tw_null      = np.random.normal(mu_null,     sigma)
Tnw_null     = np.random.normal(mu_null + 0, sigma)   # delta=0


mu_lex       = np.random.normal(300, 50, N)
a            = (0 - 0) / 50
delta_lex    = truncnorm.rvs(a=a, b=np.inf, loc=0, scale=50, size=N)
Tw_lex       = np.random.normal(mu_lex,             sigma)
Tnw_lex      = np.random.normal(mu_lex + delta_lex, sigma)


fig, axes = plt.subplots(1, 2, figsize=(12, 5), sharey=True)


axes[0].hist(Tw_null,  bins=80, alpha=0.5, color="steelblue", label="Tw (words)",
          density=True)
```

```
axes[0].hist(Tnw_null, bins=80, alpha=0.5, color="orange",    label="Tnw (non-
    words)", density=True)
axes[0].set_title("Null Model (δ = 0)")
axes[0].set_xlabel("Recognition time (ms)")
axes[0].set_ylabel("Density")
axes[0].legend()

axes[1].hist(Tw_lex,  bins=80, alpha=0.5, color="steelblue", label="Tw (words)",
        density=True)
axes[1].hist(Tnw_lex, bins=80, alpha=0.5, color="orange",    label="Tnw (non-
    words)", density=True)
axes[1].set_title("Lexical-Access Model (δ ~ N+(0,50))")
axes[1].set_xlabel("Recognition time (ms)")
axes[1].legend()

plt.suptitle("Prior Predictions: Null vs Lexical-Access Model", fontsize=13)
plt.tight_layout()
plt.show()

print("Null model:")
print(f"  Mean Tw:  {Tw_null.mean():.2f} ms")
print(f"  Mean Tnw: {Tnw_null.mean():.2f} ms")
print(f"  Difference: {Tnw_null.mean() - Tw_null.mean():.2f} ms")

print("\nLexical-access model:")
print(f"  Mean Tw:  {Tw_lex.mean():.2f} ms")
print(f"  Mean Tnw: {Tnw_lex.mean():.2f} ms")
print(f"  Difference: {Tnw_lex.mean() - Tw_lex.mean():.2f} ms")
```

```
url = "https://raw.githubusercontent.com/yadavhimanshu059/CGS698C/main/notes/
    Module-2/recognition.csv"
dat = pd.read_csv(url, index_col=0)
Tw  = dat["Tw"].values
Tnw = dat["Tnw"].values

N = 10000
sigma = 60


mu_null  = np.random.normal(300, 50, N)
Tw_null  = np.random.normal(mu_null, sigma)
Tnw_null = np.random.normal(mu_null, sigma)


mu_lex    = np.random.normal(300, 50, N)
a         = (0 - 0) / 50
delta_lex = truncnorm.rvs(a=a, b=np.inf, loc=0, scale=50, size=N)
Tw_lex    = np.random.normal(mu_lex,             sigma)
Tnw_lex   = np.random.normal(mu_lex + delta_lex, sigma)


fig, axes = plt.subplots(1, 2, figsize=(12, 5), sharey=True)

bins = np.linspace(-100, 800, 80)
```

```
axes[0].hist(Tw_null,  bins=bins, alpha=0.4, color="steelblue", label="Predicted
    Tw",  density=True)
axes[0].hist(Tnw_null, bins=bins, alpha=0.4, color="orange",    label="Predicted
    Tnw", density=True)
axes[0].hist(Tw,       bins=20,   alpha=0.7, color="blue",       label="Observed
    Tw",  density=True, histtype="step", linewidth=2)
axes[0].hist(Tnw,      bins=20,   alpha=0.7, color="red",        label="Observed
    Tnw", density=True, histtype="step", linewidth=2)
axes[0].set_title("Null Model vs Observed Data")
axes[0].set_xlabel("Recognition time (ms)")
axes[0].set_ylabel("Density")
axes[0].legend(fontsize=8)


axes[1].hist(Tw_lex,  bins=bins, alpha=0.4, color="steelblue", label="Predicted
    Tw",  density=True)
axes[1].hist(Tnw_lex, bins=bins, alpha=0.4, color="orange",    label="Predicted
    Tnw", density=True)
axes[1].hist(Tw,      bins=20,   alpha=0.7, color="blue",       label="Observed
    Tw",  density=True, histtype="step", linewidth=2)
axes[1].hist(Tnw,     bins=20,   alpha=0.7, color="red",        label="Observed
    Tnw", density=True, histtype="step", linewidth=2)
axes[1].set_title("Lexical-Access Model vs Observed Data")
axes[1].set_xlabel("Recognition time (ms)")
axes[1].legend(fontsize=8)

plt.suptitle("Prior Predictions vs Observed Data", fontsize=13)
plt.tight_layout()
plt.show()

print("Observed data:")
print(f"  Mean Tw:  {Tw.mean():.2f} ms")
print(f"  Mean Tnw: {Tnw.mean():.2f} ms")
print(f"  Observed difference: {Tnw.mean() - Tw.mean():.2f} ms")

print("\nLexical-access prior predicted difference: ~40 ms")
print("Null prior predicted difference: ~0 ms")
print(f"\nObserved difference ({Tnw.mean() - Tw.mean():.2f} ms) is much closer to
     Null model's prediction (0 ms)")


url = "https://raw.githubusercontent.com/yadavhimanshu059/CGS698C/main/notes/
    Module-2/recognition.csv"
dat = pd.read_csv(url, index_col=0)
Tw  = dat["Tw"].values
Tnw = dat["Tnw"].values

sigma = 60


delta_grid = np.linspace(0, 200, 200)
mu_grid    = np.linspace(100, 600, 150)
dmu        = mu_grid[1] - mu_grid[0]
```

```
a = (0 - 0) / 50

post_delta = []

for delta in delta_grid:


    integral_mu = 0.0
    for mu in mu_grid:
        L_Tw       = np.prod(norm.pdf(Tw,  loc=mu,         scale=sigma))
        L_Tnw      = np.prod(norm.pdf(Tnw, loc=mu + delta, scale=sigma))
        p_mu       = norm.pdf(mu, loc=300, scale=50)
        integral_mu += L_Tw * L_Tnw * p_mu * dmu


    p_delta = truncnorm.pdf(delta, a=a, b=np.inf, loc=0, scale=50)

    post_delta.append(integral_mu * p_delta)

post_delta = np.array(post_delta)

map_delta = delta_grid[np.argmax(post_delta)]
print(f"MAP estimate of delta: {map_delta:.2f} ms")


plt.figure(figsize=(8, 5))
plt.plot(delta_grid, post_delta, color="steelblue", linewidth=2)
plt.fill_between(delta_grid, post_delta, alpha=0.3, color="steelblue")
plt.axvline(map_delta, color="red", linestyle="--", label=f"MAP δ = {map_delta:.1
    f} ms")
plt.xlabel("δ (ms)")
plt.ylabel("Unnormalized posterior density")
plt.title("Unnormalized Posterior of δ  Lexical-Access Model")
plt.legend()
plt.tight_layout()
plt.show()
```